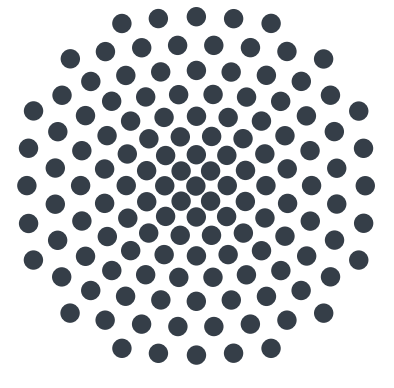
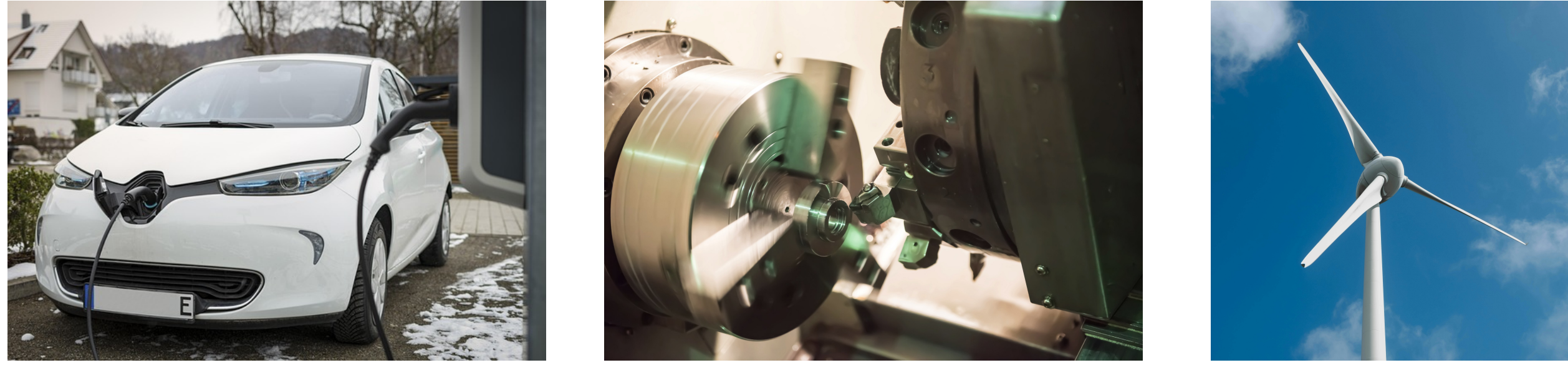


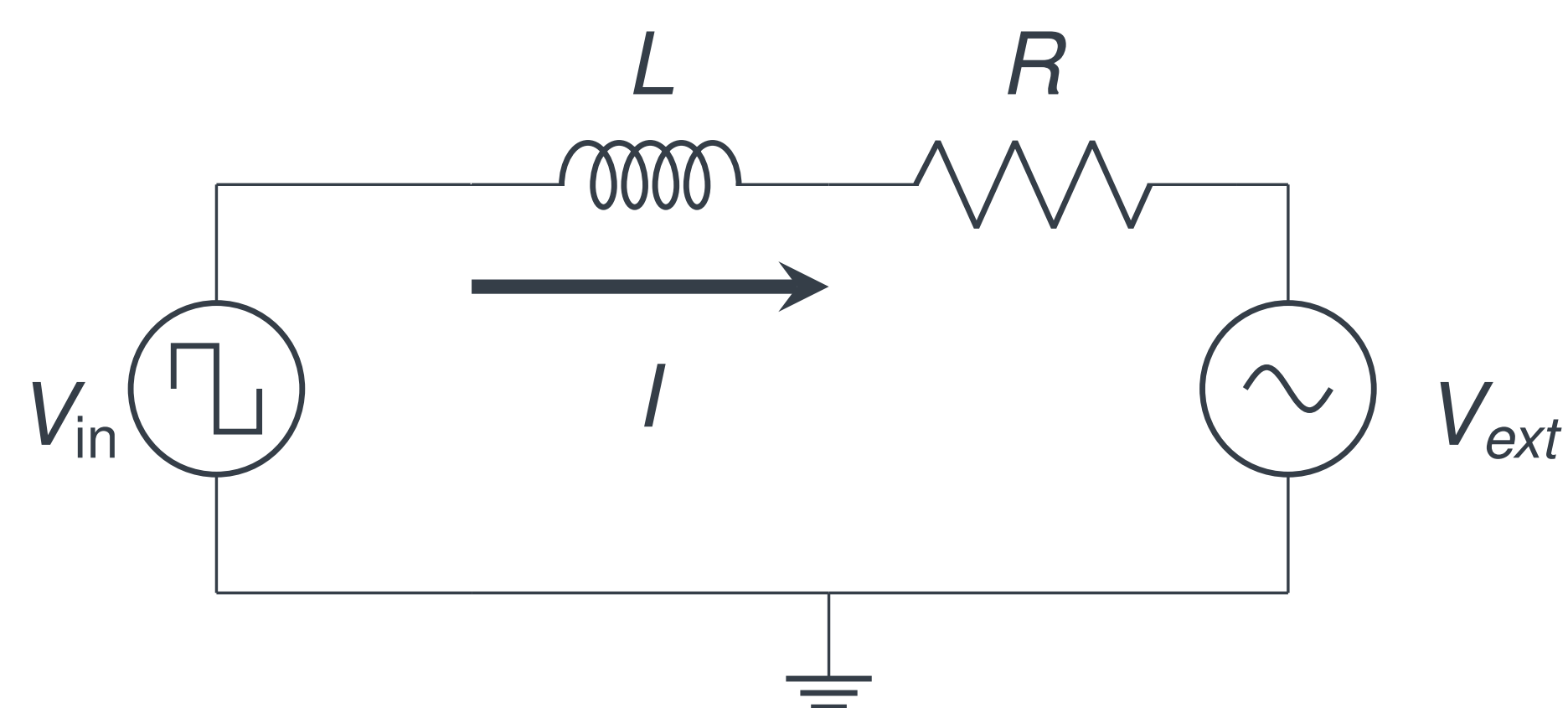


Motivation

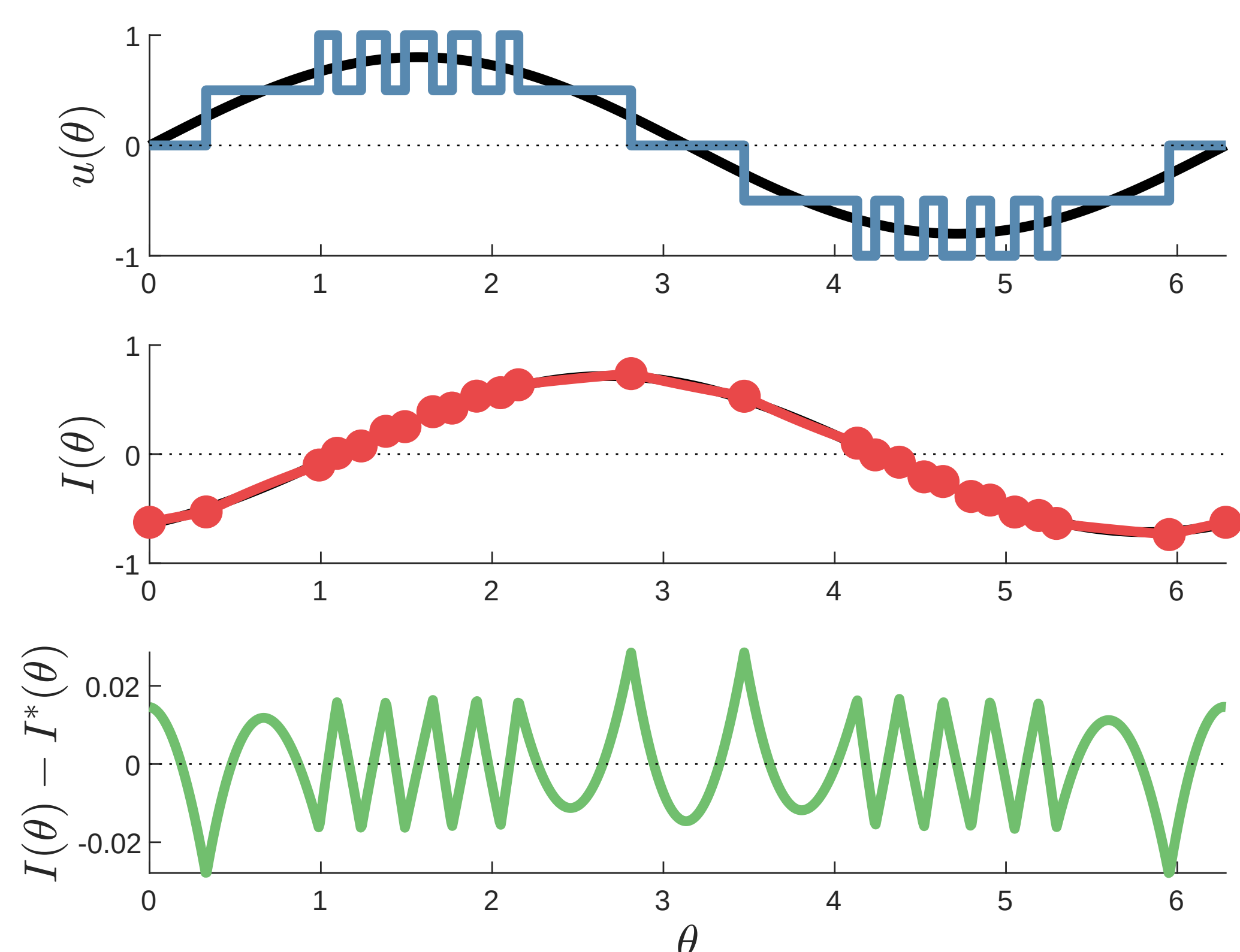
Electric drive systems are a core technology in industry and modern life.



These systems must be strictly controlled in order to produce desired motor speed, torque, current, and magnetic flux responses.



DC/AC power converters (inverters) are a component in motor control. Inverters can switch between a finite number of voltage levels to produce a load current I . The goal of this work is to bound the minimum possible energy draw $\|I\|_2^2$, which is monotonically related to the tracking error $\|I - I^*\|_2^2$ and the current demand distortion of I .



Suboptimality 2.151×10^{-5} in $\|I\|_2^2$, with 24 switches and $R/L = 0.5$

Minimizing these objectives improves energy efficiency, increases component lifetime, maintains power quality, and reduces economic costs.

Optimal Pulse Patterns

This minimization is an instance of a *nonconvex* Optimal Pulse Pattern problem [1]: choose a sequence $u(\theta)$ with levels u^i and angles α^i :

$$u(\theta) = \begin{cases} u^0 & \theta \in [0, \alpha^1) \\ u^i & \theta \in [\alpha^i, \alpha^{i+1}) \\ u^k & \theta \in [\alpha^k, 2\pi) \end{cases}$$

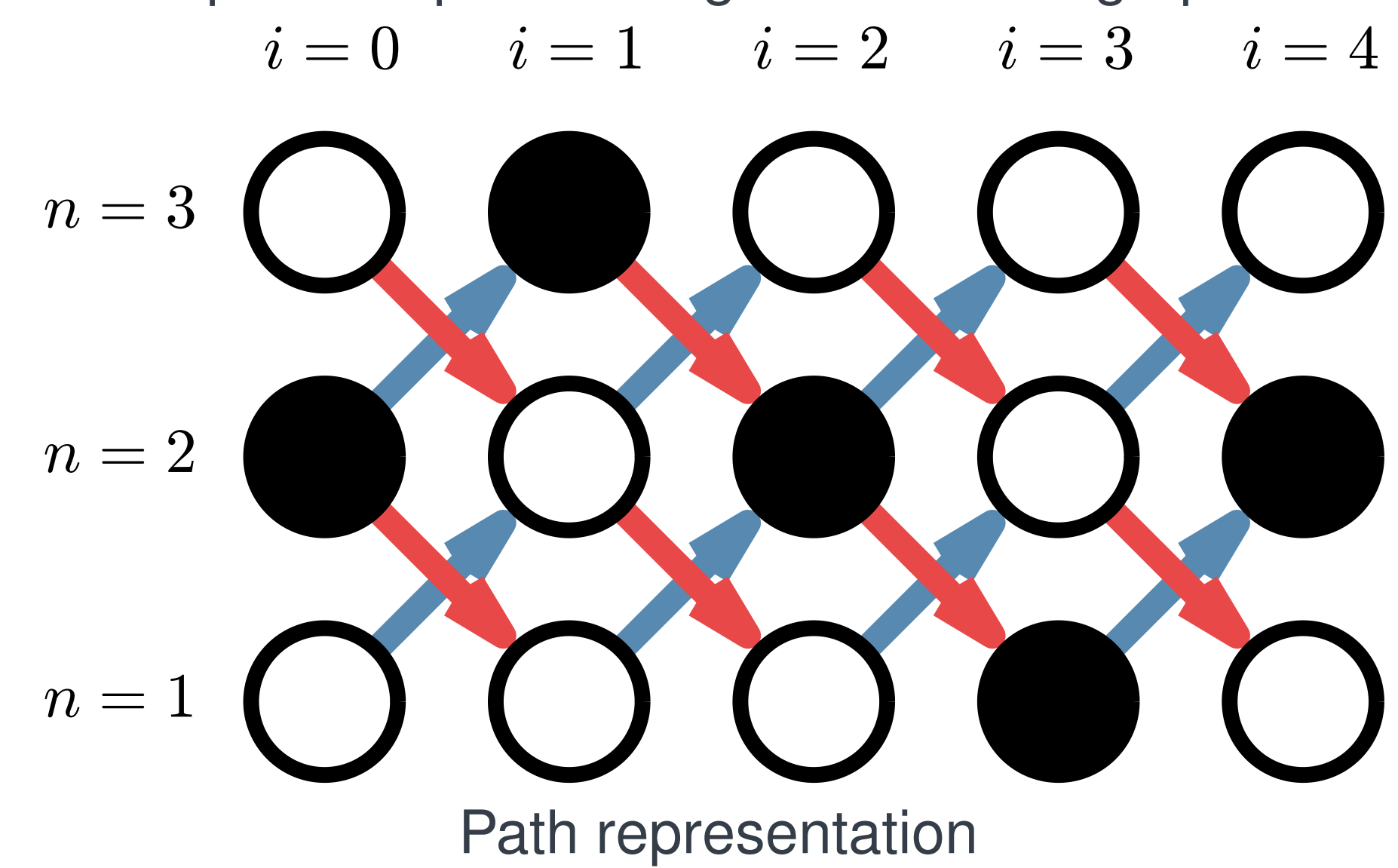
Possible (nonconvex $\star$) constraints/considerations include

- Switching Frequency
- Spacing Between Switches
- Harmonics Specifications $\star$
- Symmetries
- Switching Restrictions $\star$
- Power Budget $\star$

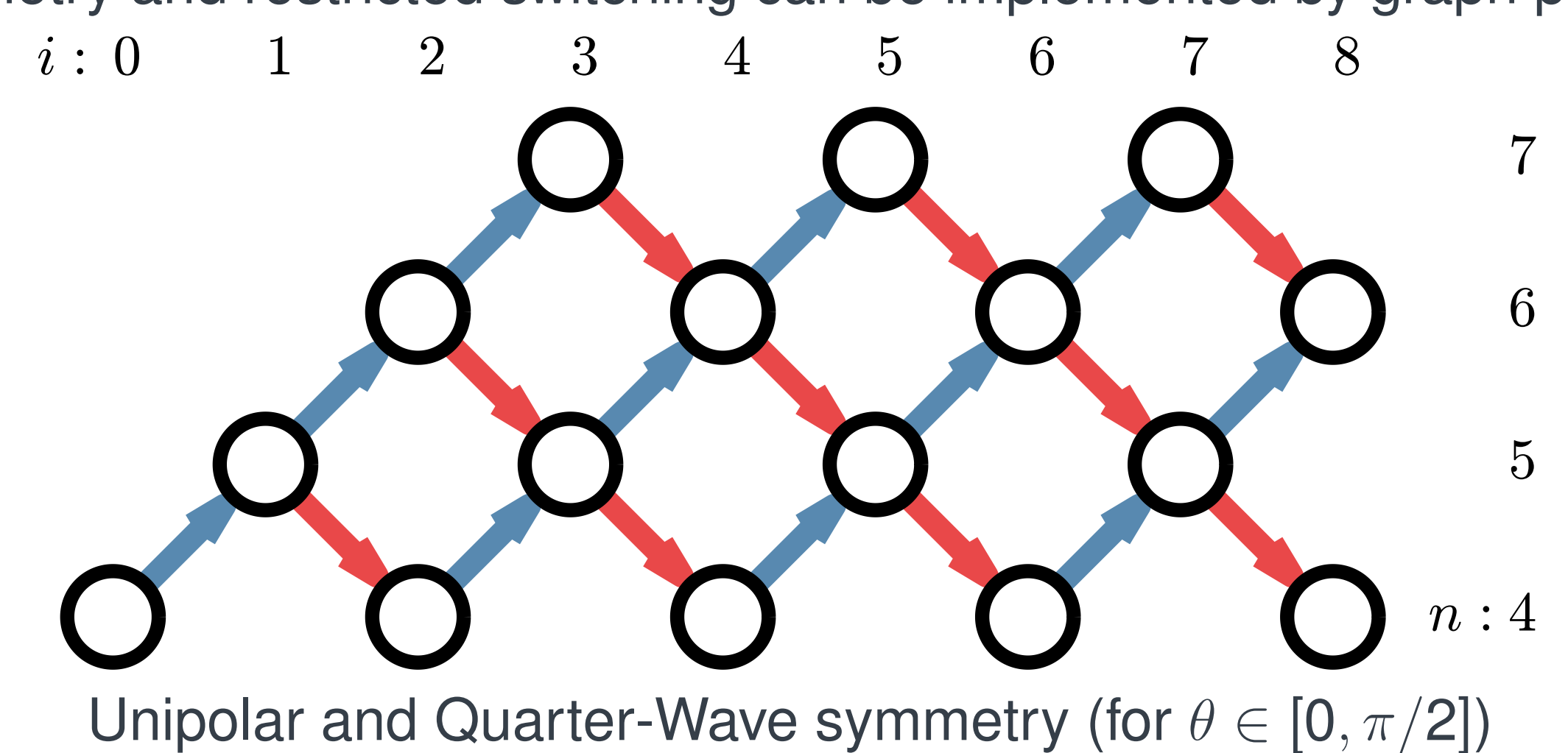
Other methods for pattern design include Selective Harmonics Elimination, Carrier/Space Vector Modulation, Finite-set MPC, and MP³C.

Optimal Pulse Patterns as Optimal Control

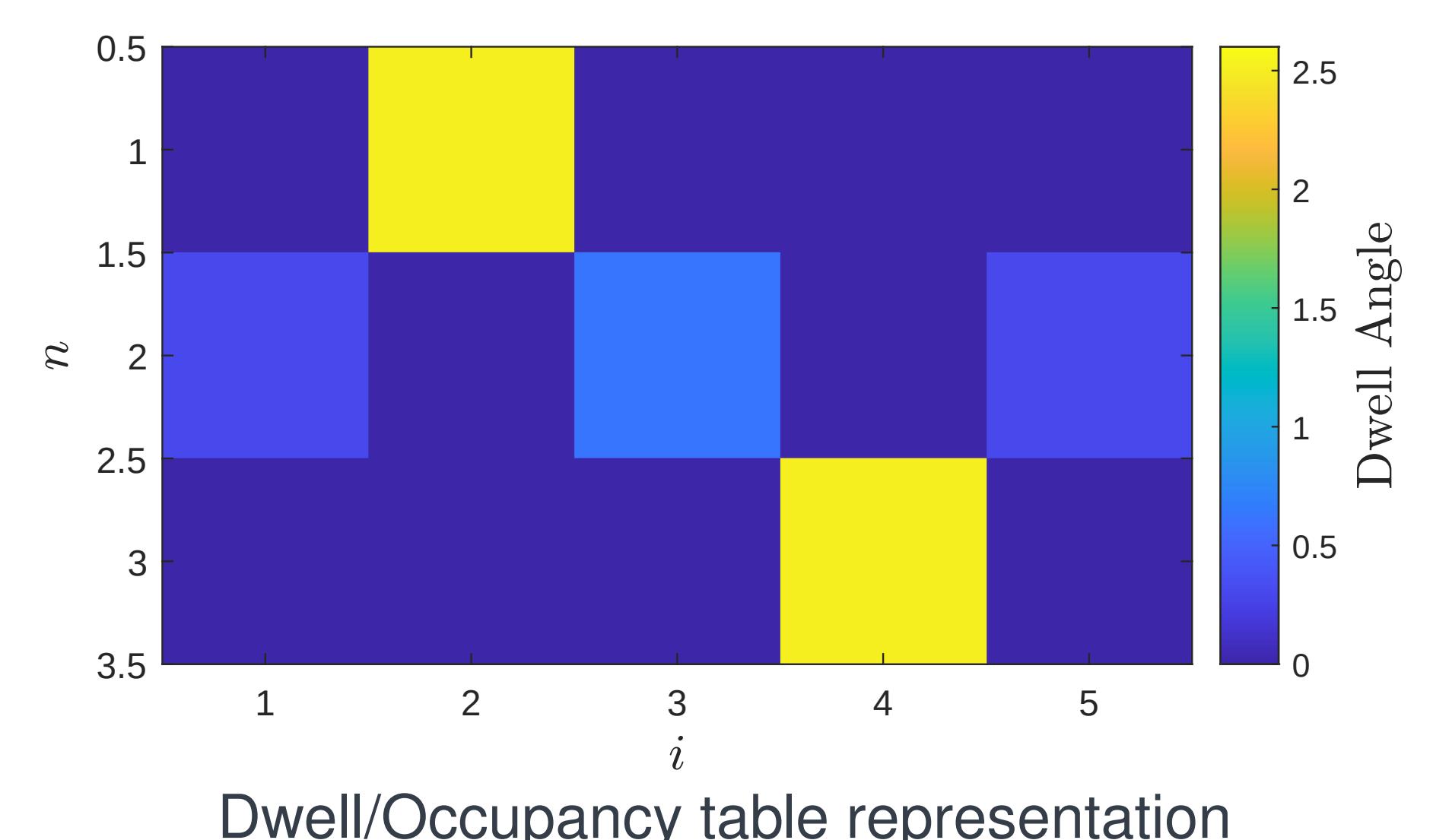
We convert Optimal Pulse Patterns into a mode-selecting periodic Optimal Control Problem in a hybrid system. The modulation levels $\{u^i\}$ are converted into a periodic path through a transition graph



Symmetry and restricted switching can be implemented by graph pruning



The angle differences $\alpha^{i+1} - \alpha^i$ correspond to the time taken at a mode



The tracked dynamics involve the angle θ , clock angle since last jump ϕ , and load current I_L . Per-mode and jump dynamics within $\theta \in [0, 2\pi]$ are

$$\text{when in mode: } \begin{bmatrix} \dot{\theta} \\ \dot{\phi} \\ \dot{I} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ u^i - \frac{R}{L}I \end{bmatrix} \quad \text{upon leaving mode: } \begin{bmatrix} \theta^+ \\ \phi^+ \\ I_L^+ \end{bmatrix} = \begin{bmatrix} \theta \\ 0 \\ I \end{bmatrix}$$

The mode-selecting optimal control reformulation remains nonconvex. We then use existing convex relaxation techniques in optimal control to lower-bound the energy cost [2]. Solving a sequence of semidefinite programs in rising polynomial degree gives increasingly accurate lower bounds to the minimal energy draw in a single-phase electric drive. Future work will include three-phase considerations.

Cooperations

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- Tobias Geyer (ABB Industrial Drives, Switzerland)

References

- [1] Giuseppe S Buja. Optimum Output Waveforms in PWM Inverters. *IEEE Trans. Ind. Appl.*, (6):830–836, 1980.
- [2] Pengcheng Zhao, Shankar Mohan, and Ram Vasudevan. Optimal control of polynomial hybrid systems via convex relaxations. *IEEE Transactions on Automatic Control*, 65(5):2062–2077, 2020.